From Episodes to Abstractions

Latent Hierarchical Memory in 1,908 AI Conversations

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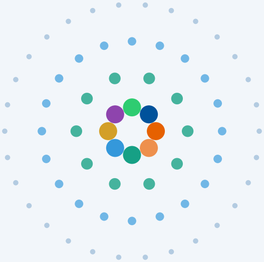


- outer: 1,908 episodes
- 500 meta-concepts
- 50 themes
- inner: 8 domains

github.com/queelius/chatgpt-complex-net

ISCS 2026 – La Rochelle, France

The Problem: Archives Are Flat. Knowledge Is Not.

What you have

Chronological list

- Bayesian inference
- Cooking pasta
- Solomonoff induction
- R package debugging
- Philosophy of mind
- ⋮

Structure is invisible.

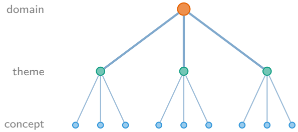
extract → concepts

The Question

Does **structure** emerge from a long archive of AI conversations — the way it does in human memory?

What's hidden

Latent hierarchy



If it's there, we should find it.

Two Ideas from Cognitive Science

Hippocampal
fast, episodic
specific events

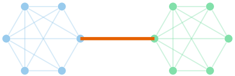
consolidate ⟷

Neocortical
slow, statistical
extracts regularities

Prediction:
New conversations reuse *existing* concepts more and more — vocabulary growth **slows**

[McClelland et al. 1995; Kumaran et al. 2016]

And: human semantic memory is a small-world network



Dense local clusters + a few global shortcuts.
Roget's, WordNet, word associations:
small-worldness ≈ 5–15

[Steyvers & Tenenbaum 2005]

The Hypothesis

If AI conversation archives are an *externalised* semantic memory, we should see both signatures: **sublinear vocabulary growth** *and* **small-world topology**.

Pipeline: Concepts → Embeddings → Hierarchy

Raw episodes
1,908 conversations
Dec 2022 – Apr 2025

→

Extract concepts
LLM (Claude)
3–10 per episode

→

Embed
each concept
→ 768-d vector

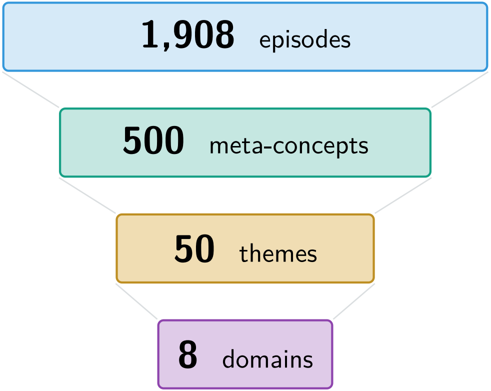
→

Cluster
hierarchical, 4-level
episode → . . . → domain

Key instruction in the prompt: *"Be specific. Say 'Metropolis–Hastings algorithm,' not 'statistics.'"*

Output: a graph linking each episode to its concepts, then grouped into a four-level hierarchy.

The Reveal: A Four-Level Hierarchy Emerges



Ward linkage on concept embeddings,
cut at 3 heights · ∼3.5× compression per level

Eight interpretable domains

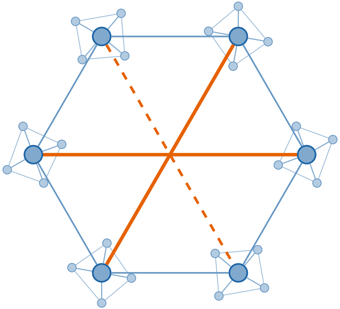
Software Engineering	1,074
Math & Optimisation	634
Statistical Methods	510
Philosophy & AI Theory	489
ML & Network Science	274
DevOps & Integration	273
AI Safety & Research	177
LLM Engineering	86

Two surprises

- Domains are **easy to name** — but there's no clean number of clusters.
- Most episodes **belong to several domains at once**.

Signature 1: Small-World Topology

Concept co-occurrence network
499 nodes, 5,935 edges



Compared to a random graph

[Humphries & Gurney 2008]

Local clustering	**6.89**×
Shortest paths	1.05×
Small-worldness	**6.6**

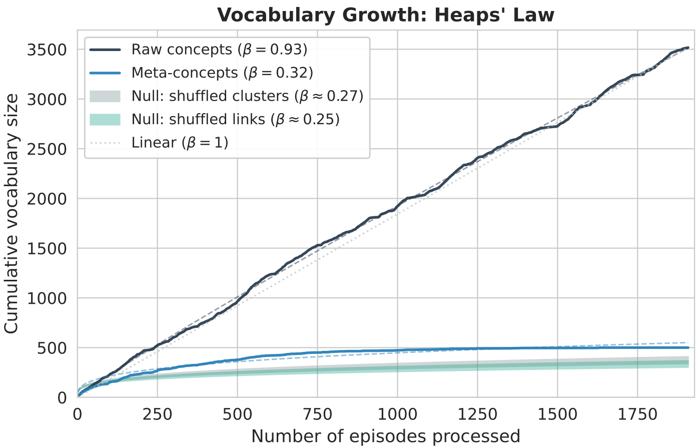
Matches human memory

(small-worldness)

Roget's Thesaurus	≈ 13
WordNet	≈ 15
Word associations	≈ 5.6
Our archive	**≈ 6.6**

[comparison values: Steyvers & Tenenbaum 2005]

Signature 2: Sublinear Vocabulary Growth



Heaps' law: $V(n) = K \cdot n^{\beta}$

[Heaps 1978]

Raw concepts	$\beta = 0.93$
Meta-concepts	$\boldsymbol{\beta = 0.32}$

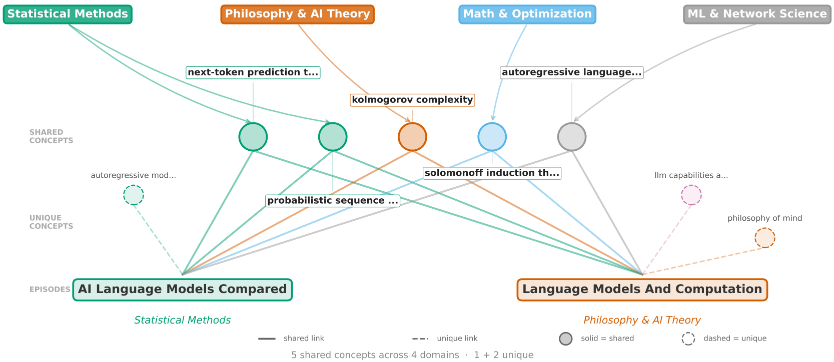
Caveat: with *k* fixed at 500

Some saturation is automatic. So we compare against **null models**.

Two nulls, both reject

- Shuffled clusters: $\beta = 0.27$
- Shuffled links: $\beta = 0.25$
- Both $p < 10^{-3}$

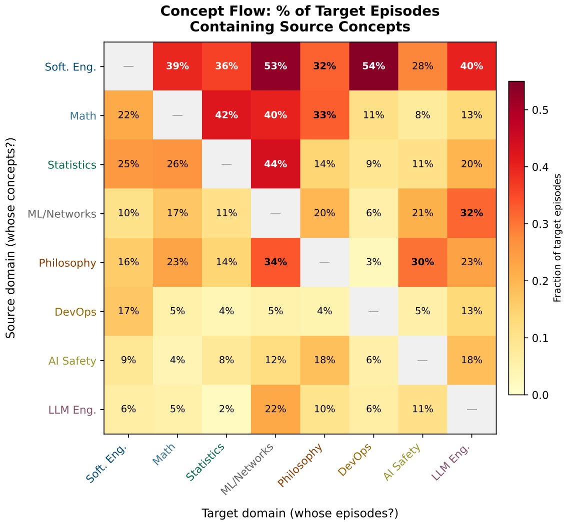
77% of Episodes Span Multiple Domains



One-label methods *(Louvain, k-means, BERTopic)*: each episode gets one cluster. These two end up apart — connection vanishes.

Set-based methods: each episode is a *set* of concepts. Two episodes share concepts across *four* domains and stay linked.

Who Borrows From Whom? (Concept Flow)



Two questions per domain

Where do its concepts go? (rows)

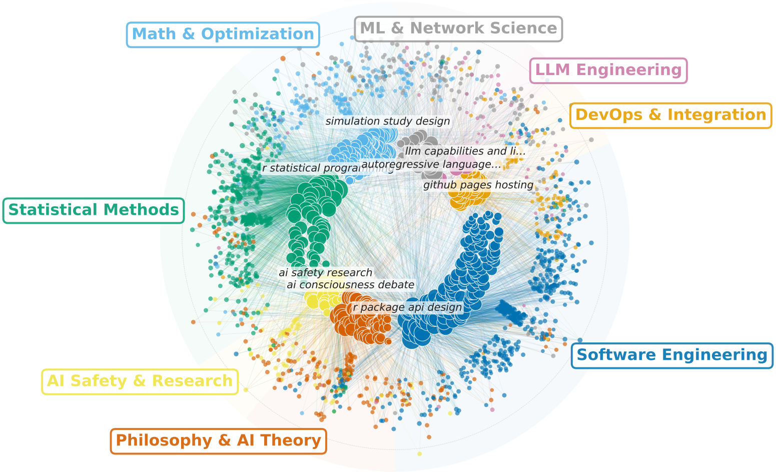
Which concepts come in? (columns)

What the heatmap says

- On average **39% below** random
  $\Rightarrow$ domains really are separate
- LLM Eng. $\to$ ML/Net: **2.67**$\times$ random
  $\Rightarrow$ specific dependencies

Software Eng. broadcasts to all 7 others.

The Full Picture



Inner: 499 concepts (coloured by domain). Outer: 1,908 episodes. Edges: episode→concept.

Takeaways & What's Next

What we found

- 4-level hierarchy $1{,}908 \to 500 \to 50 \to 8$
- Small-world structure matches human memory
- Vocabulary saturates as CLS predicts
- 77% span ≥ 2 domains

Why it matters

- Archives = **externalised semantic memory**
- Natural index for **concept-level Graph RAG**
- Knowledge is a **continuum, not a partition**

Limits & next

- **N=1** single user. Replicate on WildChat
- No natural k
- Extend to **agentic** sessions

Treat AI conversation archives as **structured cognitive artifacts** — the structure was there all along; we just needed the right lens.

Questions? queelius/chatgpt-complex-net

Backup: Method Details

Concept extraction

- Model: `claude-3-5-sonnet-20241022`, T=0
- Per episode: first 20 messages, 500-char cap
- Target: 3–10 noun phrases, 2–5 words
- 20 parallel agents (no shared vocabulary)
- Post-hoc dedup: geometric (no string matching)

Embedding

- Model: `nomic-embed-text`
- 768-dim, 8k context, L2-normalised

Hierarchical clustering

- Ward linkage, Euclidean distance
- Cuts at $k = 500, 50, 8$
- Silhouette monotonic for $k \geq 20$ (no natural boundaries)

Yield

- 7,555 raw concept mentions
- 3,517 unique (after case dedup)
- 500 meta-concepts after Ward
- 7,153 bipartite edges
- 5,935 co-occurrence edges on 499 nodes

Backup: Robustness to k

k	Heaps' β	Small-worldness	Clustering C	Edges
50	0.061	1.1	0.830	906
100	0.103	1.4	0.652	2,286
200	0.158	2.0	0.425	4,111
300	0.225	2.9	0.351	5,073
500	**0.320**	**6.57**	**0.329**	**5,935**
700	0.382	12.6	0.351	6,593
1000	0.467	25.9	0.401	7,291

Qualitative conclusions (small-worldness > 1, sublinear β, null rejection) hold for every k tested.
Quantitative numbers shift smoothly: the 500-cluster cut is a useful summary, not a privileged scale.